



Alex Towell

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Alex Towell

PhD Student | AI/ML Research | Mathematics

About Me PhD student in Computer Science focused on reliable and secure ML systems. My current work studies adversarial robustness in ransomware prevention; in controlled experiments our approach achieved a 10^{-8} false-negative rate. I prefer careful, incremental engineering, clear writing, and reproducible results. I build data and backend systems, contribute to open-source tools, and am especially interested in AI safety, evaluation, and interpretability.

Education

2024 - Present, Southern Illinois University

PhD in Computer Science

- Research Focus: AI Safety, Alignment, Mechanistic Interpretability, Adversarial Robustness
- Advisor: Dr. Hiroshi Fujinoki
- Expected Graduation: 2028

2021 - 2023, Southern Illinois University-Edwardsville

MS in Mathematics

- GPA: 3.9/4.0
- Concentration: Statistics and Operations Research
- Awarded R.N. Pendergrass Graduate Scholarship

2012 - 2015, Southern Illinois University-Edwardsville

MS in Computer Science

- GPA: 4.0/4.0
- Concentration: Artificial Intelligence
- Awarded Outstanding Graduate Student (School of Engineering)

2006 - 2010, Southern Illinois University-Edwardsville

BS in Computer Science

- GPA: 3.6/4.0
- Awarded Outstanding Junior (Dept. of Computer Science)

Technical Skills

Languages

Python (primary), R, C/C++ , Java, TypeScript, Bash, LaTeX

AI/ML

PyTorch, HuggingFace Transformers, CUDA, LLM fine-tuning, NumPy, Pandas

Systems

Linux, Docker, Git, PostgreSQL, Elasticsearch, Redis, REST APIs

Open Source

Contributor/maintainer; code review; documentation; small tooling



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Publications

- H. Fujinoki, **A. Towell**, V.A. Thota, “Preventing Ransomware Damages using In-Operation Off-Site Backup to Achieve a 10^{-8} False-Negative Miss-Detection Rate,” *International Conference on Computing and Intelligence (ICCI)*, 2025
- **A. Towell**, “Reliability Estimation in Series Systems: Maximum Likelihood Techniques for Right-Censored and Masked Failure Data,” Master’s Project, Department of Mathematics and Statistics, Southern Illinois University Edwardsville, 2023
- **A. Towell**, H. Fujinoki, “Estimating How Confidential Encrypted Searches Are Using Moving Average Bootstrap Method,” *IEEE International Conference on Cloud Computing Technology and Science (CloudCom)*, 2016
- **A. Towell**, “Encrypted Search: Enabling Standard Information Retrieval Techniques Using New Secure Index Types,” Master’s Thesis, ProQuest Dissertations & Theses, 2015

Selected Research & Projects

AlgoTree

Python utilities for tree-like data (FlatForest/TreeNode) with a CLI (jt) to query and manipulate hierarchical data; used for inspecting nested structures.

Elasticsearch-LM

Prototype fine-tuning to translate natural language into Elasticsearch Query DSL using transformers and synthetic data; evaluates translation accuracy and failure modes.

Reverse-Process Synthetic Data

Exploratory algorithms to generate math-reasoning training data; emphasis on data quality, clear assumptions, and reproducibility.

Oblivious Computation

Explored space–security trade-offs for privacy-preserving computation; built small prototypes for workload partitioning.

IDOT D8 Infrastructure System

Lead developer for a traffic-monitoring system; proposed and planning multimodal models to assess incidents and camera quality. Designed for reliability, auditability, and straightforward operations.



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Academic Experience

2024–Present, Research Assistant, SIU

- Cybersecurity research with Dr. Hiroshi Fujinoki on ransomware detection using probabilistic data structures and careful evaluation
- Built prototypes for telemetry analysis and automated response with LLMs; documented limits and failure cases
- Lead developer for Illinois Department of Transportation District 8 (IDOT D8) application with planned multimodal AI components
- Co-authored study reporting 10^{-8} false-negative miss-detection under controlled conditions (ICCI 2025)

2021–2022, Teaching Assistant, SIUE

- Taught lab courses in mathematics and statistics

2012–2015, Research/Graduate/Teaching Assistant, SIUE

- Supported research in AI and machine learning while pursuing MS in Computer Science
- Taught undergraduate computer science lab sessions
- Developed encrypted search algorithms for confidential data retrieval systems